

Autonomous Microstructure Alpha Discovery under False-Discovery Control

Order-Book Imbalance, Conditional ICs, and the Adverse-Selection Barrier in Hyperliquid Perps

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Thesis

The gap between statistical predictability and economic exploitability, measured rather than asserted. 236 engineered features, 100 ms cadence, BTC/ETH/SOL perps, horizons 1 s–15 min. **The central result is negative**, and it is the reason the whole programme pivoted to execution.

The headline pair

$$IC_{\text{uncond}} \approx 0.45 \rightarrow IC_{\text{cond}} \approx 0.03$$

condition on states where a directionally-correct passive (mid-cross) fill occurs

Holds on all three symbols. This is a *structural adverse-selection barrier*, not an estimation artifact: the states in which you get filled are systematically the states in which you were wrong. **Execution, not signal quality, is the binding constraint.**

Signal structure

- Order-book imbalance dominates directionally; $IC \approx 0.45$ at 1–5 s, decaying to ≈ 0 by 5 min.
- Same feature hierarchy across BTC/ETH/SOL — universality, not a per-symbol fit.
- Clean directional/volatility orthogonality (developed fully in #7).
- Robustness: 7 of 8 checks pass for BTC, 6 of 8 for ETH and SOL.

The five-gate replication protocol

Every candidate passes all five or lands in a documented graveyard *with its rejection reason*.

G1 Discovery — rank IC over an adaptive threshold *and* incremental $dIC \geq 0.05$ beyond already-registered signals.

G2 Cost — gross return/trade ≥ 0.1 bps. Kills statistically real signals too small to pay for themselves.

G3 Temporal — IC holds on ≥ 1 further date (White-2000 reality check in spirit).

G4 Cross-symbol — IC holds on ≥ 1 other symbol.

G5 Correlation dedup — $\rho < 0.7$ against every registered signal. Enforces breadth, not just strength.

Multiple-testing machinery

BH-FDR at $q = 0.05$ across all tests each cycle: order $p_{(1)} \leq \dots \leq p_{(m)}$, take $j^* = \max\{j : p_{(j)} \leq jq/m\}$. Threshold *adapts* as the registry strengthens:

$$\min IC(t) = \max(\text{floor}, \text{median}\{IC_i : i \in R(t)\} \times 0.8)$$

plus continuous decay monitoring on registered signals. The apparatus exists to suppress the multiple-testing bias that pervades empirical asset pricing — a discovery platform that tests thousands of hypotheses *needs* this or it manufactures winners.

The hierarchical combiner

Three layers: slow directional bias gates fast microstructure timing. Recovers positive cost-adjusted OOS IC of 0.18/0.25/0.36 (BTC/ETH/SOL). [!] On a **thin window** — the paper says so, and it is the one number here to distrust. The five algorithms that carried this claim were all REFUTED by the Q4 alpha-skeptic kill gate 2026-07-30 (wrong-venue cost tier + sweep-harness defects). Re-check before P1 camera-ready; the conditional-IC result above is unaffected.

What is actually new

1. A reproducible methodology for FDR-controlled microstructure research (the apparatus is the contribution, not any one signal).
2. Direct evidence that a strong, universal, robust predictor can be *economically inert* under realistic passive execution.

Known and not re-claimed: that imbalance/OFI predicts short-horizon returns (Cont–Kukanov–Stoikov 2014, deep-LOB literature).

Why the negative result is the valuable one

It converts an open-ended search (“find a better signal”) into a bounded engineering problem (“get filled without adverse selection”), and it explains why every naive backtest on this data looks spectacular and every honest one does not. Everything downstream — maker execution, GAP-04/HF1/A4, the conditional-IC gate as the D1 go/no-go — descends from this section.

Siblings

#7 sharpens this into a measurement-first paper (and adds: conditioning on *any* fill *raises* IC to 0.52, which isolates adverse selection rather than decay) · #6 generalises the gate discipline to a typed analytical unit · #5 asks what an exogenous signal adds to a book bounded this way · #8 is the venue instrument built on the conclusion.

Data & apparatus

Hyperliquid BTC/ETH/SOL at 100 ms; Rust ingestor on the L2 WS feed $\rightarrow$ 236-dim vector/symbol/100 ms $\rightarrow$ columnar storage. Discovery window 2026-05-19..21 (3 full days, ≈ 2.17 M ticks/symbol); robustness spans **31 dates 04-19..06-04, of which 23 are sufficiently complete** — the gap is itself a data-quality statement. [!] Of 236 schema features only **207 are populated**; the rest depend on a position-attribution pipeline outside the paper’s scope. Quote 207, not 236, when describing what was actually tested. Fifteen feature categories: raw book state, imbalance, trade flow, realized/range volatility, permutation & tick entropy, exchange context (funding, OI), trend & Hurst, illiquidity (Kyle, Amihud, Roll), toxicity (VPIN, effective/realized spread), composites.

The loop

Ingest (Rust) $\rightarrow$ Discover (agents) $\rightarrow$
Replicate (5 gates) $\rightarrow$ Deploy (paper)

with a feedback path that *retires* decayed signals. Four agents at different timescales — tick, minute-to-hour, hour-to-day, and a cross-agent orchestrator. The apparatus, not any signal, is the contribution.

The eight checks

Seven confirmatory, the eighth decisive. Per-day IC stable across 23 valid dates; present 24/7 with no dead zone (dip 12–16 UTC, strongest 04–08 UTC); above 0.34 in *both* volatility regimes on all symbols under a median split — **slightly stronger in low vol**, i.e. imbalance is more informative when the book is not being disrupted. The eighth check is the conditional IC, and it is the one that fails.

Reading order for a supervisor

#2 (this) for the apparatus + the barrier $\rightarrow$ #7 for the measurement census $\rightarrow$ #6 for the theory of breadth. #2 and #7 cross-cite; do not send both without saying which is the lead.